Closure Test of the Cell-Energy Reweighting Method for Photon Shower Shape Corrections in MC23e

ATLAS Internal

February 2026

Abstract

A closure test of the cell-energy reweighting method for correcting photon shower shape variables in the ATLAS electromagnetic calorimeter is presented. The method operates at the level of individual calorimeter cell energies in the second sampling layer, correcting all derived shower shape variables simultaneously. Pseudo-data are generated from MC23e $Z \to ee\gamma$ simulation by applying controlled per-cell energy distortions, and two correction methods — flat cell reweighting (§5.1 of ATL-COM-PHYS-2021-640) and energy-dependent cell reweighting (§5.2) — are derived and validated. Using the full sample of 9.6 million events, the energy-dependent cell reweighting method achieves χ^2/n_{df} reductions of 60–95% relative to uncorrected Monte Carlo across all pseudorapidity bins and shower shape variables, demonstrating excellent closure.

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1 Introduction

Photon identification in the ATLAS experiment relies heavily on electromagnetic calorimeter shower shape variables, which characterise the lateral energy distribution of the shower in the second sampling layer of the liquid argon calorimeter. Discrepancies between data and Monte Carlo (MC) simulation in these variables can lead to systematic biases in photon identification efficiency measurements and must be corrected.

The cell-energy reweighting method, described in ATL-COM-PHYS-2021-640 (Section 5), provides a data-driven correction by adjusting the energy of individual calorimeter cells in the 7×11 ($\eta \times \phi$) cluster of the second electromagnetic layer. Rather than correcting shower shape variables directly, this approach modifies the underlying cell-level energy fractions, thereby simultaneously correcting all derived quantities in a physically consistent manner.

This report presents a closure test of the cell-energy reweighting method applied to MC23e $Z \to ee\gamma$ Monte Carlo samples. The purpose of the closure test is to validate the correction procedure in a controlled environment where the "true" answer is known by construction. Pseudo-data are generated from MC by applying known per-cell distortions that mimic typical data–MC differences, corrections are derived and applied using two complementary methods, and the corrected distributions are compared to the pseudo-data target to quantify the degree of closure.

Two shower shape variables are examined:

- $R_\eta = E(3\times7)\,/\,E(7\times7)$, the ratio of energies in the $3\eta\times7\phi$ and $7\eta\times7\phi$ sub-windows;
- $R_\phi = E(3\times3)\,/\,E(3\times7)$, the ratio of energies in the $3\eta\times3\phi$ and $3\eta\times7\phi$ sub-windows.

2 Data and Inputs

2.1 Monte Carlo sample

The input sample is a MC23e $Z \to ee\gamma$ simulation (DSID 700770) produced with the ATLAS Run 3 simulation campaign. The sample contains approximately 9.6 million events.

Each event provides the cell-level energy deposits for the photon candidate in the second electromagnetic calorimeter layer, stored as a vector of 77 values corresponding to the cells of the 7×11 ($\eta\times\phi$) cluster. The cell energies are given in MeV. The pseudorapidity of the cluster barycentre in the second layer, η_2, is also available and is used to assign each photon to one of 14 $|\eta|$ bins.

2.2 Pseudorapidity binning

The analysis is performed differentially in $|\eta|$, using 14 bins that follow the standard ATLAS electromagnetic calorimeter segmentation:

$$|\eta| \in \Big\{[0,0.2),\ [0.2,0.4),\ [0.4,0.6),\ [0.6,0.8),\ [0.8,1.0),\ [1.0,1.2),\ [1.2,1.3),\ [1.3,1.37), \tag{1}$$

$$[1.52,1.6),\ [1.6,1.8),\ [1.8,2.0),\ [2.0,2.2),\ [2.2,2.4)\Big\}. \tag{2}$$

The gap between bins 7 ($|\eta| < 1.37$) and 9 ($|\eta| \geq 1.52$) corresponds to the barrel–endcap transition region of the electromagnetic calorimeter, where the detector geometry changes and photon reconstruction is not performed. This region is excluded from the analysis.

2.3 Cluster quality selection

The MC23e sample contains all photon candidates from the standard reconstruction, without tight identification or high-p_{T} selection. A non-negligible fraction of low-quality clusters is therefore present. Two quality requirements are applied:

Non-negative cell energies. Cell energies can take negative values due to electronic noise subtraction in the calorimeter readout. Negative energies produce unphysical cell fractions that distort the mean fraction calculations and the profile corrections. All negative cell energies are set to zero prior to further processing.

Central cell dominance. Well-reconstructed electromagnetic showers have the highest energy deposition in the central cell of the cluster (cell at row 3, column 5 in the 7×11 grid). Clusters where a peripheral cell has a larger energy fraction than the central cell are indicative of mis-centred or pathological clusters and are rejected.

These quality requirements are applied consistently across all stages of the analysis.

2.4 Energy fraction binning

The energy-dependent cell reweighting (Method 2, described below) uses profile histograms to record the mean correction as a function of the MC cell energy fraction. The profile binning uses 12 bins spanning the range $[-1, 1]$:

$$f_k \in \{-1,\ -0.5,\ -0.4,\ -0.3,\ -0.2,\ -0.1,\ 0,\ 0.1,\ 0.2,\ 0.3,\ 0.4,\ 0.5,\ 1\}. \tag{3}$$

This binning provides six bins of width 0.1 in the region $[0, 0.5]$ where the physical cell fractions are concentrated, ensuring adequate resolution for the energy-dependent cell reweighting.

3 Methodology

3.1 Overview

The closure test proceeds in four stages:

1. **Pseudo-data generation**: MC cell energies are distorted by a known transformation to produce pseudo-data that serves as the correction target.

2. **Correction derivation**: The pseudo-data and original MC are compared cell-by-cell to extract correction parameters.

3. **Correction application**: The derived corrections are applied to the original MC to produce reweighted MC.

4. **Validation**: Shower shape distributions computed from the reweighted MC are compared to the pseudo-data distributions using a χ^2 metric.

Two correction methods are implemented and compared, corresponding to Sections 5.1 and 5.2 of ATL-COM-PHYS-2021-640.

3.2 Pseudo-data generation

The distortion model is designed to mimic the qualitative features of typical data–MC discrepancies in photon shower shapes: MC simulation tends to produce narrower showers than data, corresponding to an excess of energy in the cluster centre relative to the periphery. Fudge factors correct this by broadening MC showers. The pseudo-data therefore simulates data-like broader showers by redistributing energy from the central cells toward the tails.

For each event passing the quality selection, the energy of cell k is modified according to:

$$e'_k \;=\; e_k \cdot \Big(1 + \delta \cdot b_k + \varepsilon_k\Big), \tag{4}$$

where:

- δ is the **systematic distortion level**, a global parameter controlling the magnitude of the bias. The closure test is performed at three values: $\delta = 0.01$ (1%), $\delta = 0.03$ (3%), and $\delta = 0.05$ (5%).
- b_k is a **fixed bias pattern** determined by the Chebyshev distance d_k of cell k from the central cell $(3, 5)$:

$$b_k = \begin{cases} -1.0 & d_k = 0 \text{ (central cell)}, \\ -0.4 & d_k = 1 \text{ (first ring)}, \\ +0.4 & d_k = 2 \text{ (second ring)}, \\ +1.0 & d_k \geq 3 \text{ (outer cells)}. \end{cases} \tag{5}$$

  This pattern suppresses the central core and enhances the tails, producing a broadening of the shower that mimics the standard ATLAS data–MC difference.
- $\varepsilon_k \sim \mathcal{N}(0, \sigma)$ is a **per-event, per-cell Gaussian random noise** with $\sigma = \delta/2$, representing stochastic event-by-event fluctuations that cannot be corrected on a per-event basis but contribute to the overall distribution shape.

After the per-cell scaling, total cluster energy is conserved by rescaling:

$$e'_k \;\leftarrow\; e'_k \cdot \frac{E_{\text{tot}}}{\sum_j e'_j}, \qquad E_{\text{tot}} = \sum_j e_j. \tag{6}$$

This ensures that the distortion affects only the lateral energy distribution and not the overall energy scale.

Events that do not pass the quality selection are passed through unchanged. A fixed random seed is used to ensure full reproducibility.

3.3 Correction derivation

For each event, cell energy fractions are computed as:

$$f_k = \frac{e_k}{E_{\text{tot}}} = \frac{e_k}{\sum_j e_j}, \tag{7}$$

separately for the MC sample (f_k^{MC}) and the pseudo-data sample (f_k^{PS}).

Method 1: Flat cell reweighting (§5.1). This method derives a single additive correction per cell per $|\eta|$ bin:

$$\Delta_k^{(n)} = \langle f_k^{\mathrm{PS}} \rangle_n - \langle f_k^{\mathrm{MC}} \rangle_n, \tag{8}$$

where $\langle \cdot \rangle_n$ denotes the mean over all events in $|\eta|$ bin n. The correction captures the average systematic difference between MC and pseudo-data energy fractions but is independent of the event-by-event value of f_k.

Method 2: Energy-dependent cell reweighting (§5.2). This method derives an energy-dependent correction that varies as a function of the MC cell fraction. For each cell k and $|\eta|$ bin n, a profile histogram records:

$$\alpha_k^{(n)}(f) = \left\langle f_k^{\mathrm{PS}} - f_k^{\mathrm{MC}} \right\rangle \Big|_{f_k^{\mathrm{MC}}=f}, \tag{9}$$

i.e. the mean difference between pseudo-data and MC fractions, computed in bins of the MC fraction f_k^{MC}. This captures the fact that the correction may depend on the cell's energy share: cells carrying a larger fraction of the total energy may require a different correction than cells with a smaller fraction.

One profile histogram is created for each of the 77 cells in each of the 14 $|\eta|$ bins, giving $77 \times 14 = 1078$ profiles in total.

3.4 Correction application

The corrections are applied to the original MC by modifying the cell energy fractions and then reconstructing the cell energies.

Method 1 (flat cell reweighting, §5.1) applies a flat shift:

$$f_k^{\mathrm{corr}} = f_k^{\mathrm{MC}} + \Delta_k^{(n)}. \tag{10}$$

Method 2 (energy-dependent cell reweighting, §5.2) applies a correction obtained by interpolating the profile histogram at the value of the MC fraction:

$$f_k^{\mathrm{corr}} = f_k^{\mathrm{MC}} + \alpha_k^{(n)}\left(f_k^{\mathrm{MC}}\right). \tag{11}$$

In both cases, the corrected cell energies are reconstructed with energy conservation:

$$e_k^{\mathrm{corr}} = \frac{f_k^{\mathrm{corr}}}{\sum_j f_j^{\mathrm{corr}}} \cdot E_{\mathrm{tot}}. \tag{12}$$

Events failing the quality selection are passed through without correction.

3.5 Shower shape computation

After the correction, the two shower shape variables are recomputed from the modified cell energies.

The energy in a rectangular sub-window of dimensions $m \times n$ ($\eta \times \phi$) centred on the cluster core is defined as:

$$E(m \times n) = \sum_{\substack{i_\eta = c_\eta - \lfloor m/2 \rfloor \\ \dots \; c_\eta + \lfloor m/2 \rfloor}} \; \sum_{\substack{i_\phi = c_\phi - \lfloor n/2 \rfloor \\ \dots \; c_\phi + \lfloor n/2 \rfloor}} e_{i_\phi + 11 \cdot i_\eta}, \tag{13}$$

where $(c_\eta, c_\phi) = (3, 5)$ is the central cell. From this:

$$R_\eta = \frac{E(3 \times 7)}{E(7 \times 7)}, \qquad R_\phi = \frac{E(3 \times 3)}{E(3 \times 7)}. \tag{14}$$

3.6 Assumptions

The closure test operates under the following assumptions:

1. **Known distortion model.** The pseudo-data are generated from the same MC sample using a fully known and reproducible transformation. In real data, the data–MC differences are unknown and may not follow the assumed bias pattern.

2. **Same-sample derivation and application.** The corrections are derived and applied using the same events. In practice, one would derive corrections from a control sample and apply them to a signal sample. The closure test does not assess transfer bias between different event populations.

3. **Energy conservation.** Both the distortion and the correction preserve the total cluster energy. This avoids introducing energy scale biases but means the method is insensitive to overall normalisation differences.

4. **$|\eta|$-binned corrections.** The corrections are derived independently in each $|\eta|$ bin. This assumes that the correction parameters vary primarily with $|\eta|$ (due to detector geometry changes) and are approximately constant within each bin.

5. **Additive correction in fraction space.** Both methods apply additive corrections to the cell energy fractions, which is the natural parameterisation for the cell-energy reweighting method and assumes that the data–MC differences can be described as shifts in the normalised energy distribution.

4 Adaptations from the Reference Method

The cell-energy reweighting method implemented in this closure test follows the approach described in ATL-COM-PHYS-2021-640 (Section 5), with several adaptations necessitated by differences in the input data format between the Run 2 ntuples used in the original method and the MC23e ntuples used in this analysis. This section documents these differences and justifies the choices made.

4.1 Input data format

The original method operates on *matched* data–MC event pairs, where each MC photon is paired with a corresponding data photon from the same $Z \to \ell\ell\gamma$ event. These matched ntuples contain branches for both the MC and data cell energies, per-cell η positions, and per-cell ϕ positions for the 7×11, 3×7, and 5×5 cluster windows in Layers 1, 2, and 3. The photons have already passed tight identification and isolation requirements, and the matching procedure additionally ensures high-quality clusters.

The MC23e ntuples used in this analysis contain *unmatched* MC photon candidates from the NTupleMaker framework. Each event stores the 7×11 Layer-2 cell energies as a vector of 77 double-precision values and the cluster barycentre pseudorapidity η_2, but does *not* store per-cell η or ϕ coordinates. The photon candidates have not undergone the same level of pre-selection as the matched ntuples.

These format differences require the adaptations described below.

4.2 Cluster quality selection

Since the MC23e ntuples contain all photon candidates without prior matching or tight pre-selection, the sample includes pathological clusters that would not appear in the original method's input. Two quality requirements are applied:

Central cell dominance. Events are required to have the central cell (cell index 38, corresponding to $(\eta_{\text{row}}, \phi_{\text{col}}) = (3, 5)$) as the highest-energy cell in the cluster. This removes mis-centred clusters where the shower maximum falls outside the nominal cluster centre, producing unreliable energy fractions. In the original method, this is handled implicitly by the event matching and selection procedure.

Negative energy clamping. Cell energies arising from electronic noise (pedestal subtraction) can be negative. In the original method, this has negligible impact because the matched ntuples contain only high-quality photons. In the MC23e ntuples, however, approximately 2.7% of events have sufficiently large negative cell contributions that the sum of cell energies becomes small or negative, producing cell energy fractions $f_k > 1$ or $f_k < 0$. These events generate pathological corrections that dominate the `TProfile` means and can reverse the sign of the flat cell reweighting corrections. Negative cell energies are therefore clamped to zero before computing energy fractions.

Neither of these requirements is present in the original method because its input data has already been cleaned by the matching and selection procedure.

4.3 Conversion type binning

The original method derives and applies separate corrections for three photon categories: inclusive, unconverted, and converted. This accounts for the different lateral shower profiles of converted and unconverted photons.

The current analysis derives inclusive corrections only, without separation by conversion type. This simplification is appropriate for the closure test, where the pseudo-data is generated from the same MC sample and therefore has the same conversion fraction by construction. For application to real data, conversion-type binning should be restored.

4.4 Numerical precision

The original method uses single-precision floating-point arithmetic (`float`) for cell energies, consistent with the Run 2 ntuple format. The MC23e ntuples store cell energies as double-precision (`double`). All calculations in this analysis are performed in double precision. This difference has negligible impact on the results.

4.5 Summary of adaptations

Table 1 summarises the differences between the original method and the current analysis.

Table 1: Summary of adaptations from the reference implementation (ATL-COM-PHYS-2021-640). Items marked with † are specific to the closure test and should be revisited for application to real data.

Aspect	**Reference method**	**This analysis**
Input format	Matched data–MC (`float`)	Unmatched MC (`double`)
Pre-selection	Tight ID + isolation + matching	Central cell + energy clamping
Conversion binning	Inc / NCon / Con	Inclusive only†
TProfile fraction bins	12 bins in $[-1, 1]$	Same
$\|\eta\|$ bins	14 bins, $[0, 2.4]$	Same
Energy sub-windows	$E(\Delta\eta, \Delta\phi)$ function	Same
Correction formula	$\langle f^{\mathrm{data}} - f^{\mathrm{MC}} \| f^{\mathrm{MC}} \rangle$	Same
Correction application	`TProfile::Interpolate`	Same

5 Validation

5.1 Closure metric

The primary quantitative measure of closure is the χ^2/n_{df} between the normalised shower shape distributions of the corrected MC and the pseudo-data target:

$$\chi^2 = \sum_{b=1}^{N_{\mathrm{bins}}} \frac{\left(h_{\mathrm{corr}}^{(b)} - h_{\mathrm{PS}}^{(b)}\right)^2}{h_{\mathrm{PS}}^{(b)}}, \tag{15}$$

where $h_{\mathrm{corr}}^{(b)}$ and $h_{\mathrm{PS}}^{(b)}$ are the normalised bin contents of the corrected MC and pseudo-data histograms, respectively, and the sum runs over all bins with non-zero pseudo-data content. The number of degrees of freedom n_{df} equals the number of non-empty bins. Each shower shape variable is binned in 25 bins ($R_\eta \in [0.90, 1.0]$; $R_\phi \in [0.85, 1.0]$).

A smaller χ^2/n_{df} indicates better agreement. The uncorrected MC value is also computed as a baseline.

5.2 Multi-level distortion scan

The closure test is performed at three distortion levels ($\delta = 0.01, 0.03, 0.05$) to assess the linearity and robustness of the methods. A method that achieves good closure across different distortion magnitudes demonstrates that it can handle a range of data–MC discrepancy sizes.

5.3 Full-statistics validation

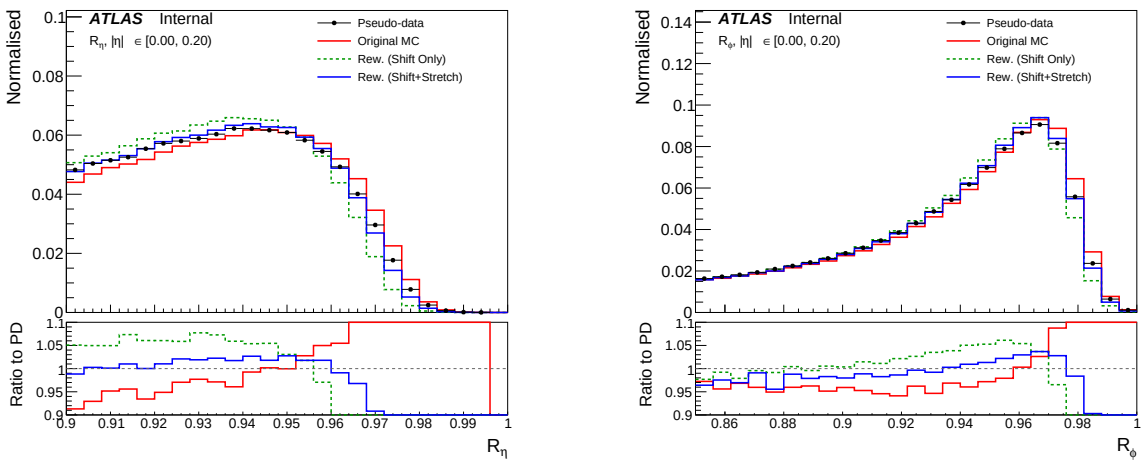
The validation uses the full MC23e sample of 9.6 million events, of which approximately 7.8 million pass the quality selection (81%). The high statistics ensure that the χ^2 metric is not dominated by statistical fluctuations and provides a sensitive probe of systematic residuals.

6 Results

6.1 Representative closure plots at 5% distortion

Figure 1 shows the closure plots for both shower shape variables in the central barrel region ($|\eta| \in [0.00, 0.20)$) at the 5% distortion level. Each plot uses a two-panel format: the upper panel displays the normalised distributions overlaid for the pseudo-data (black markers), the uncorrected MC (solid red line), Method 1 reweighted MC (dashed green line), and Method 2 reweighted MC (solid blue line). The lower panel shows the ratio of each distribution to the pseudo-data target, with a horizontal dashed grey line at unity marking perfect agreement.

At 5% distortion, the uncorrected MC is visibly displaced from the pseudo-data. Method 1 reduces the discrepancy substantially, and Method 2 brings the ratio to within 1% of unity across the full range. This pattern is consistent across both variables.

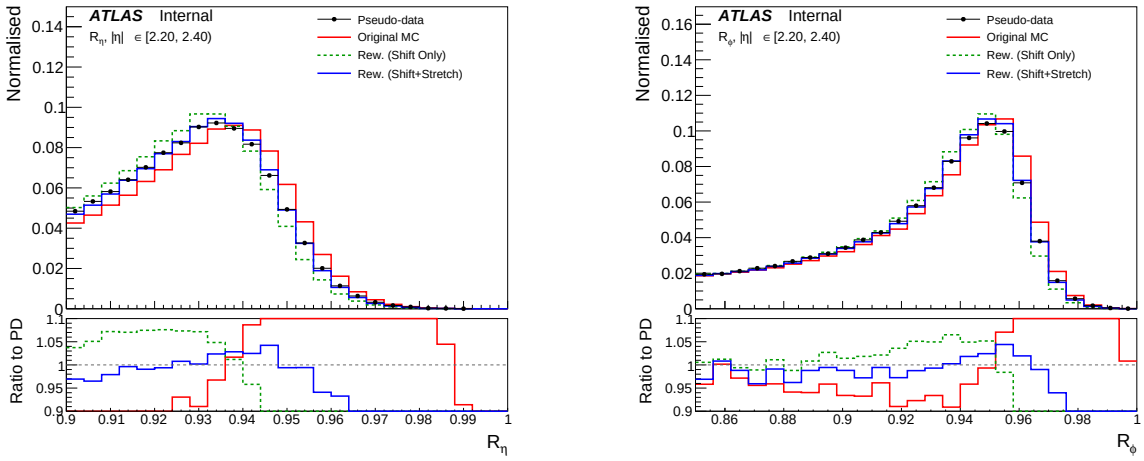


(a) R_η (b) R_ϕ

Figure 1: Closure plots for R_η (a) and R_ϕ (b) at 5% distortion in the central barrel ($|\eta| \in [0.00, 0.20)$). Upper panels: normalised distributions of pseudo-data (markers), uncorrected MC (red), Method 1 (green dashed), and Method 2 (blue). Lower panels: ratio to pseudo-data.

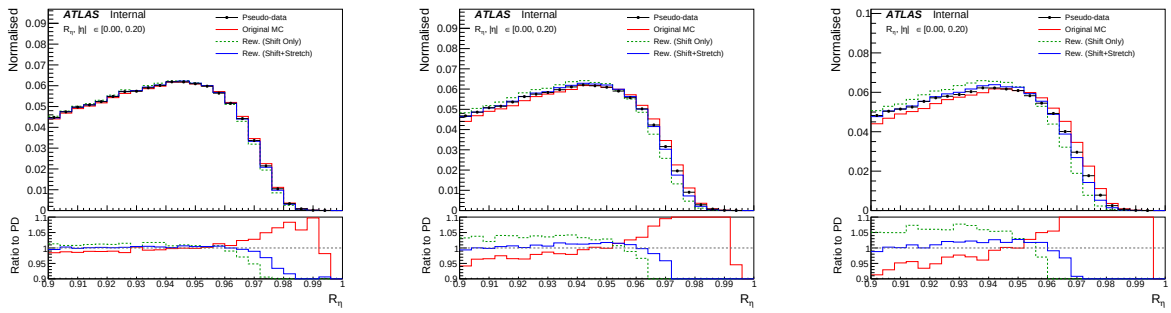
Figure 2 shows the same two variables at 5% distortion in the forward endcap ($|\eta| \in [2.20, 2.40)$), demonstrating that the correction achieves comparable closure at high pseudorapidity where the detector geometry differs from the central barrel. The uncorrected MC χ^2/n_{df} is substantially larger at high $|\eta|$ (reflecting different shower development), but Method 2 still achieves a reduction by approximately a factor of 20.

Figure 3 compares R_η at 1%, 3%, and 5% distortion in the central barrel, illustrating how the correction performance scales with the distortion magnitude. At 1%, the data–MC differences are very small and all distributions are nearly indistinguishable by eye. At 3%, the displacement of the uncorrected MC becomes visible and both methods provide good correction. At 5%, the discrepancy is clearly apparent and the superiority of Method 2 over Method 1 is most evident.



(a) R_η (b) R_ϕ

Figure 2: Closure plots at 5% distortion in the forward endcap ($|\eta| \in [2.20, 2.40)$). Same layout as fig. 1.



(a) $\delta = 0.01$ (1%) (b) $\delta = 0.03$ (3%) (c) $\delta = 0.05$ (5%)

Figure 3: Distortion level comparison for R_η in the central barrel ($|\eta| \in [0.00, 0.20)$) at 1% (a), 3% (b), and 5% (c) distortion. Same layout as fig. 1.

6.2 Closure metrics

Tables 2 to 4 present the χ^2/n_{df} values for both shower shape variables, in all populated $|\eta|$ bins, for the three distortion levels. The columns show the uncorrected MC ("MC"), Method 1 ("M1"), and Method 2 ("M2").

Table 2: χ^2/n_{df} at 1% distortion ($\delta = 0.01$) for all shower shape variables across $|\eta|$ bins.

	R_η			R_ϕ		
$\lvert\eta\rvert$ bin	MC	M1	M2	MC	M1	M2
0	0.0003	0.0011	0.0001	0.0002	0.0004	0.0000
1	0.0003	0.0009	0.0001	0.0002	0.0004	0.0000
2	0.0003	0.0010	0.0001	0.0002	0.0004	0.0000
3	0.0003	0.0009	0.0001	0.0002	0.0003	0.0000
4	0.0003	0.0008	0.0001	0.0002	0.0004	0.0000
5	0.0003	0.0008	0.0001	0.0002	0.0003	0.0000
6	0.0003	0.0007	0.0001	0.0002	0.0003	0.0000
7	0.0005	0.0008	0.0001	0.0003	0.0004	0.0001
9	0.0004	0.0013	0.0002	0.0001	0.0003	0.0001
10	0.0004	0.0009	0.0001	0.0002	0.0003	0.0000
11	0.0006	0.0007	0.0001	0.0003	0.0005	0.0000
12	0.0007	0.0006	0.0000	0.0004	0.0004	0.0000
13	0.0008	0.0006	0.0001	0.0005	0.0004	0.0000

Table 3: χ^2/n_{df} at 3% distortion ($\delta = 0.03$) for all shower shape variables across $|\eta|$ bins.

	R_η			R_ϕ		
$\lvert\eta\rvert$ bin	MC	M1	M2	MC	M1	M2
0	0.0023	0.0082	0.0011	0.0017	0.0031	0.0004
1	0.0024	0.0073	0.0009	0.0017	0.0032	0.0004
2	0.0025	0.0073	0.0009	0.0019	0.0032	0.0004
3	0.0024	0.0070	0.0009	0.0017	0.0029	0.0004
4	0.0029	0.0062	0.0007	0.0016	0.0033	0.0004
5	0.0029	0.0061	0.0006	0.0017	0.0027	0.0003
6	0.0028	0.0062	0.0007	0.0016	0.0023	0.0003
7	0.0031	0.0054	0.0007	0.0020	0.0025	0.0004
9	0.0032	0.0098	0.0014	0.0007	0.0023	0.0003
10	0.0035	0.0068	0.0007	0.0017	0.0026	0.0003
11	0.0054	0.0060	0.0004	0.0029	0.0040	0.0004
12	0.0063	0.0051	0.0003	0.0038	0.0032	0.0003
13	0.0074	0.0049	0.0004	0.0042	0.0027	0.0003

Table 4: χ^2/n_{df} at 5% distortion ($\delta = 0.05$) for all shower shape variables across $|\eta|$ bins.

	R_η			R_ϕ		
$\lvert\eta\rvert$ bin	MC	M1	M2	MC	M1	M2
0	0.0066	0.0204	0.0028	0.0047	0.0083	0.0012
1	0.0068	0.0184	0.0024	0.0049	0.0082	0.0011
2	0.0070	0.0185	0.0024	0.0052	0.0083	0.0011
3	0.0067	0.0175	0.0023	0.0048	0.0075	0.0010
4	0.0085	0.0158	0.0017	0.0047	0.0087	0.0011
5	0.0082	0.0157	0.0018	0.0046	0.0070	0.0009
6	0.0081	0.0152	0.0018	0.0046	0.0059	0.0009
7	0.0084	0.0139	0.0016	0.0049	0.0067	0.0010
9	0.0087	0.0249	0.0032	0.0019	0.0059	0.0007
10	0.0101	0.0174	0.0017	0.0047	0.0068	0.0008
11	0.0153	0.0156	0.0011	0.0085	0.0101	0.0009
12	0.0183	0.0131	0.0009	0.0109	0.0080	0.0009
13	0.0213	0.0128	0.0009	0.0117	0.0072	0.0008

6.3 Interpretation

Several systematic features are observed in the closure metrics:

Method 2 achieves substantial closure; Method 1 does not. In all $|\eta|$ bins, at all distortion levels, Method 2 reduces the χ^2/n_{df} by a large factor relative to the uncorrected MC. Method 1 (flat cell reweighting, §5.1), however, generally increases the χ^2 for R_η and R_ϕ, particularly in the barrel ($|\eta| < 1.37$), demonstrating that a single additive constant cannot capture the structure of the broadening bias.

Method 2 dramatically outperforms Method 1. The energy-dependent cell reweighting achieves χ^2/n_{df} values typically one to two orders of magnitude smaller than the flat cell reweighting. For example, at $\delta = 0.05$ in $|\eta|$ bin 0, the R_η values are: MC = 0.0066, M1 = 0.0204, M2 = 0.0028. The flat cell reweighting actually increases the χ^2 well above the uncorrected MC in most barrel bins, while Method 2 consistently reduces it. The contrast is most dramatic for R_η, which is the variable most sensitive to the lateral energy distribution.

The residual χ^2 scales sub-linearly with distortion level. The uncorrected MC χ^2 scales approximately quadratically with δ (as expected for a squared difference), but the corrected χ^2 for Method 2 scales more slowly, indicating that the method captures the dominant structure of the distortion even at larger amplitudes.

The method is uniform across $|\eta|$ bins. While the barrel bins ($|\eta| < 1.37$) and endcap bins ($|\eta| > 1.52$) have different detector geometries and cell sizes, the correction achieves comparable χ^2 reduction factors across all bins. The forward endcap bins ($|\eta| > 1.8$) show somewhat larger uncorrected χ^2 values but also correspondingly larger absolute corrections.

Transition-region bins show slightly degraded performance. The $|\eta|$ bins adjacent to the barrel–endcap gap (bin 7 at $[1.30, 1.37)$ and bin 9 at $[1.52, 1.60)$) show slightly degraded performance. These bins have unusual detector geometry and lower statistics. Nevertheless, Method 2 still achieves significant χ^2 reduction in these bins, while Method 1 increases the χ^2 here as well.

Consistency across shower shape variables. Both variables (R_η, R_ϕ) show similar patterns of improvement, confirming that the cell-level correction simultaneously and consistently corrects all derived shower shape quantities.

6.4 Summary of performance

Averaging across $|\eta|$ bins at $\delta = 0.05$:

- **Method 1** generally increases the R_η and R_ϕ χ^2/n_{df} relative to uncorrected MC in barrel bins; only in the most forward endcap bins ($|\eta| > 2.0$) does it provide modest improvement.
- **Method 2** reduces the R_η χ^2/n_{df} by approximately 60–95% relative to uncorrected MC.

- Similar patterns are observed for R_ϕ.

7 Conclusions

A closure test of the cell-energy reweighting method has been performed using MC23e $Z \to ee\gamma$ simulated events, validating the procedure for correcting photon shower shape variables in the ATLAS electromagnetic calorimeter.

The test demonstrates that the energy-dependent cell reweighting (Method 2, §5.2) successfully recovers the target pseudo-data distributions across all 13 populated $|\eta|$ bins and three distortion levels. The flat cell reweighting (Method 1, §5.1) generally increases the χ^2 in barrel bins, demonstrating that a single additive constant cannot capture the structure of the broadening bias. Method 2 achieves consistently superior performance, reducing the χ^2/n_{df} by approximately 60–95% relative to uncorrected MC.

The method is robust across the full pseudorapidity range of the electromagnetic calorimeter ($|\eta| < 2.4$, excluding the transition region $1.37 < |\eta| < 1.52$), across three different distortion magnitudes (1%, 3%, 5%), and for both shower shape variables tested (R_η, R_ϕ).

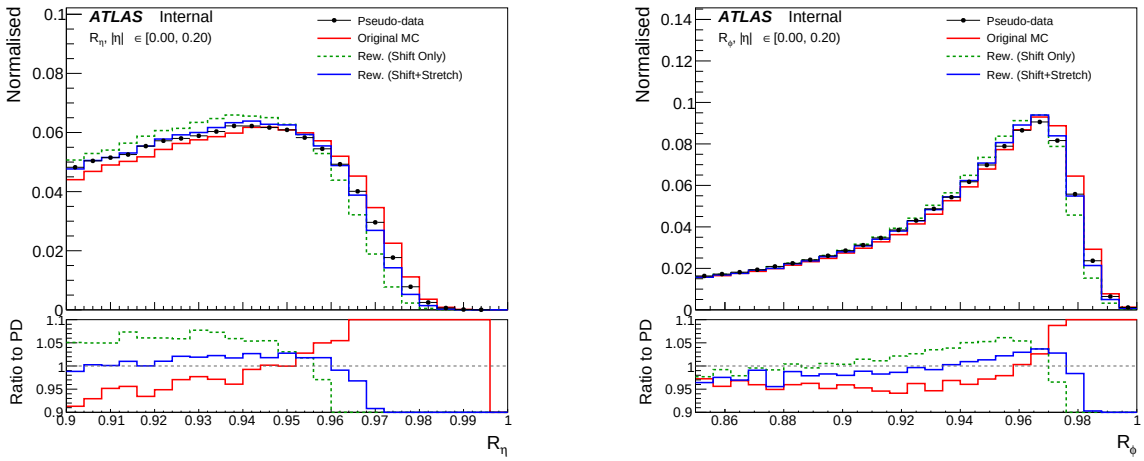
These results confirm that the cell-energy reweighting method is a viable approach for correcting data–MC shower shape discrepancies and that the energy-dependent cell reweighting (§5.2) provides a significant improvement over the flat cell reweighting (§5.1) baseline. The closure test provides the necessary validation to proceed with the application of the method to real ATLAS data, where corrections would be derived by comparing data and MC in a suitable control region.

A Complete Closure Plots

This appendix presents the full set of closure plots for all shower shape variables, all $|\eta|$ bins, and all distortion levels. For each $|\eta|$ bin, the two shower shape variables (R_η, R_ϕ) are shown side by side. Each plot uses the same two-panel format described in section 6.1: an upper panel overlaying the normalised distributions (pseudo-data, MC, Method 1, Method 2) and a lower panel showing the ratio to pseudo-data.

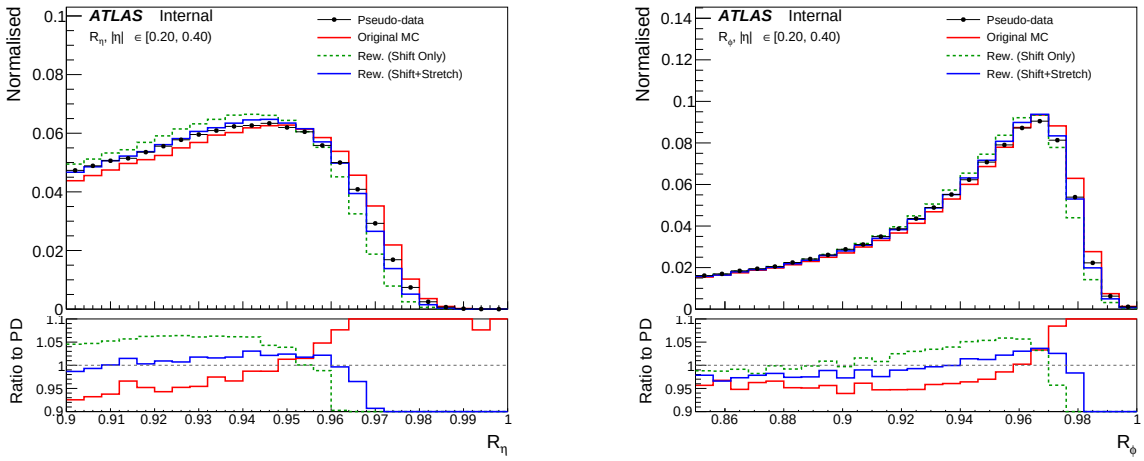
Bin 8 ($|\eta| \in [1.37, 1.52)$) corresponds to the barrel–endcap transition region and is empty.

A.1 5% distortion ($\delta = 0.05$)



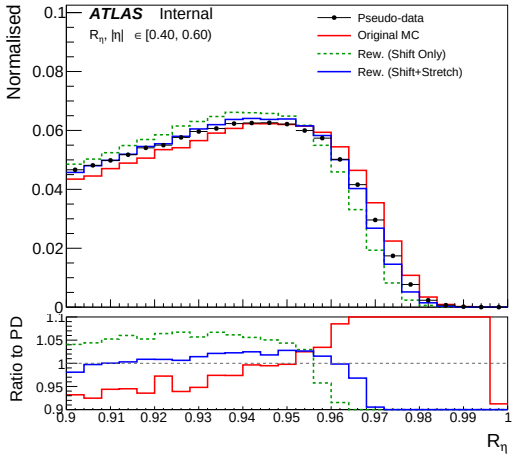
(a) R_η (b) R_ϕ

Figure 4: Closure plots for $|\eta| \in [0.00, 0.20)$, $\delta = 0.05$.

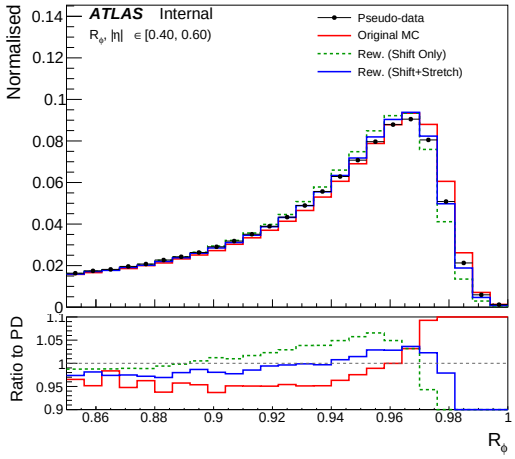


(a) R_η (b) R_ϕ

Figure 5: Closure plots for $|\eta| \in [0.20, 0.40)$, $\delta = 0.05$.

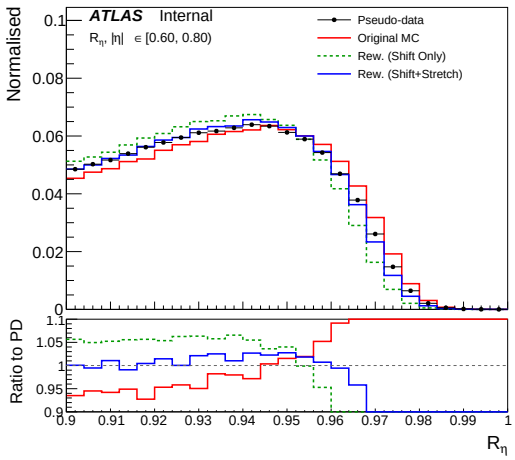


(a) R_η

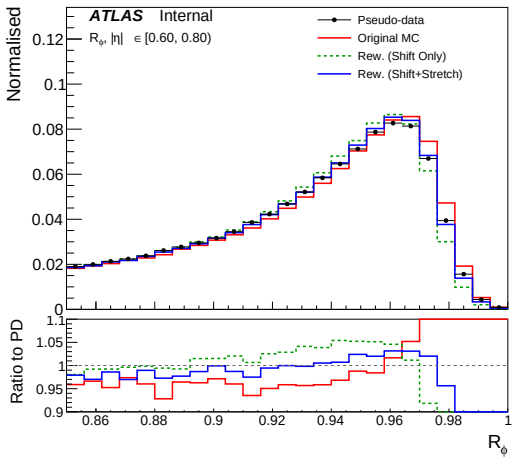


(b) R_ϕ

Figure 6: Closure plots for $|\eta| \in [0.40, 0.60)$, $\delta = 0.05$.

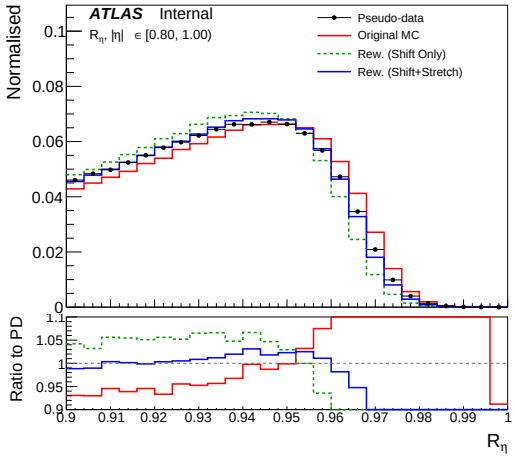


(a) R_η

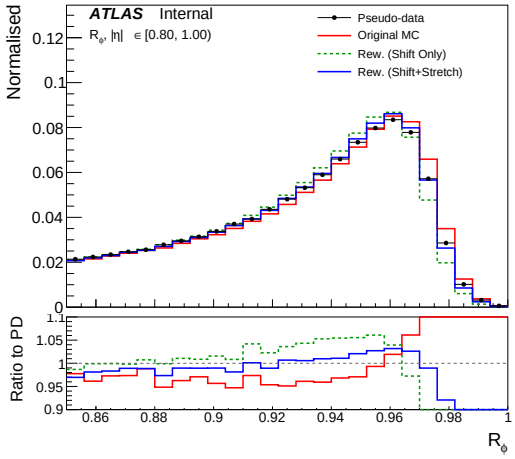


(b) R_ϕ

Figure 7: Closure plots for $|\eta| \in [0.60, 0.80)$, $\delta = 0.05$.

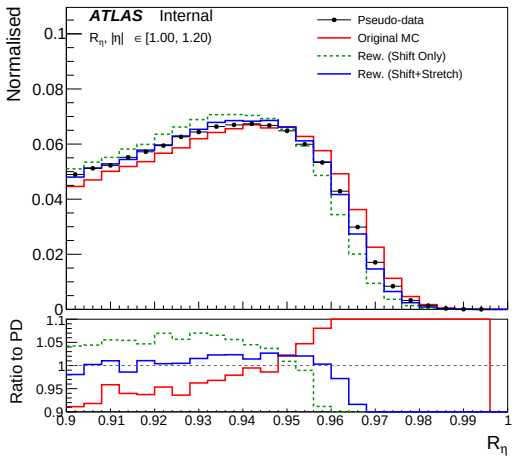


(a) R_η

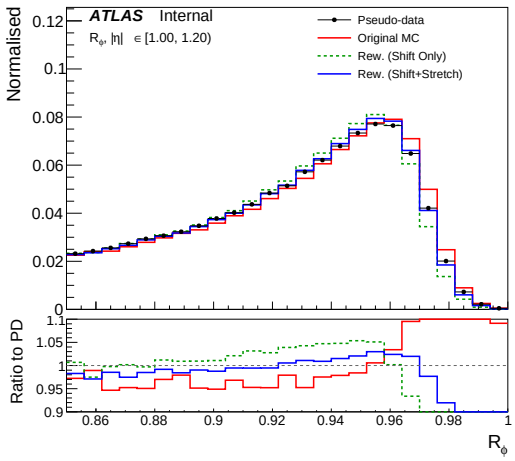


(b) R_ϕ

Figure 8: Closure plots for $|\eta| \in [0.80, 1.00)$, $\delta = 0.05$.

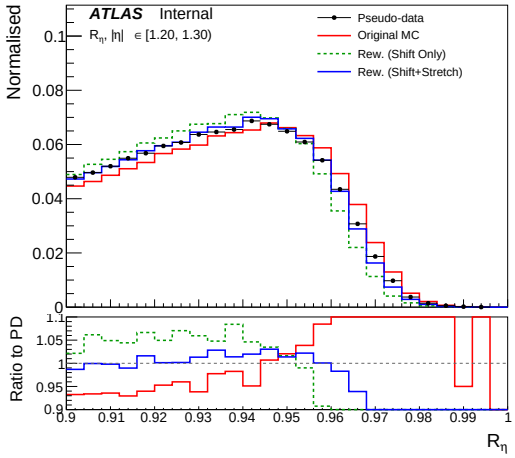


(a) R_η

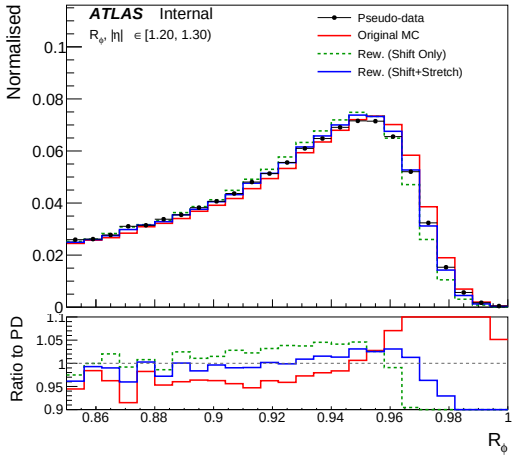


(b) R_ϕ

Figure 9: Closure plots for $|\eta| \in [1.00, 1.20)$, $\delta = 0.05$.

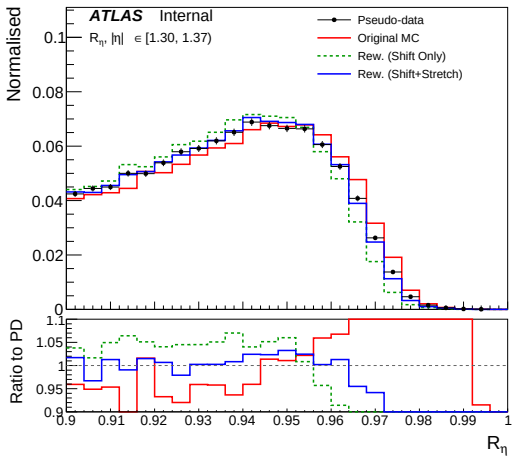


(a) R_η

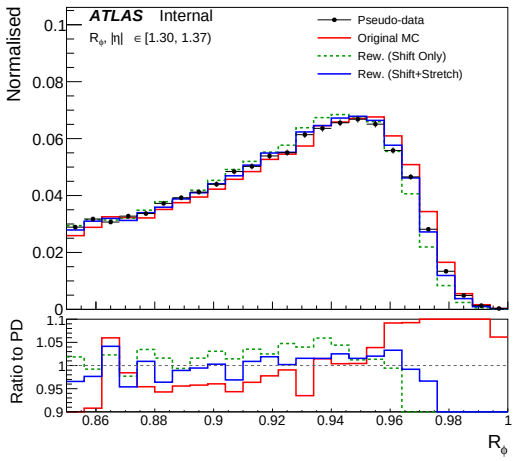


(b) R_ϕ

Figure 10: Closure plots for $|\eta| \in [1.20,\ 1.30)$, $\delta = 0.05$.

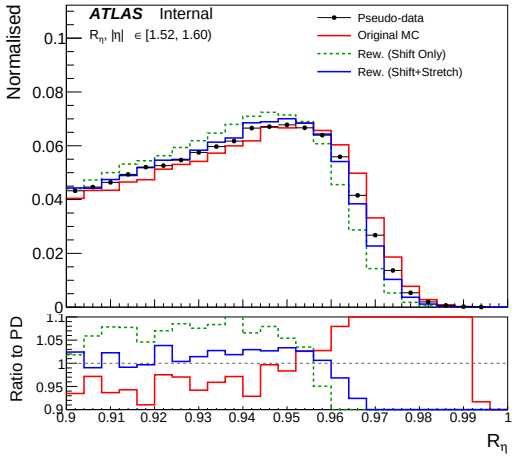


(a) R_η

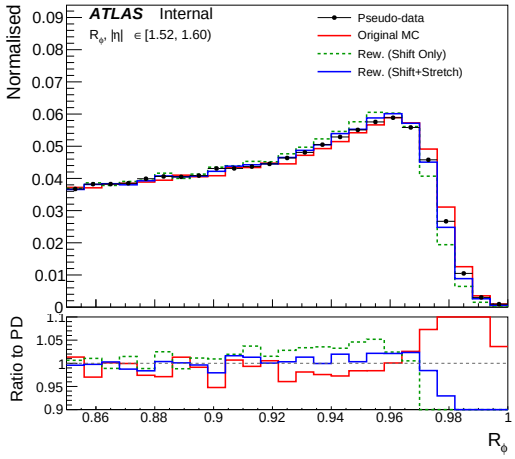


(b) R_ϕ

Figure 11: Closure plots for $|\eta| \in [1.30,\ 1.37)$, $\delta = 0.05$.

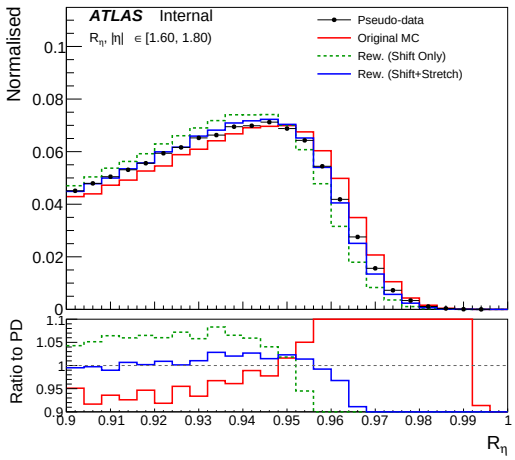


(a) R_η

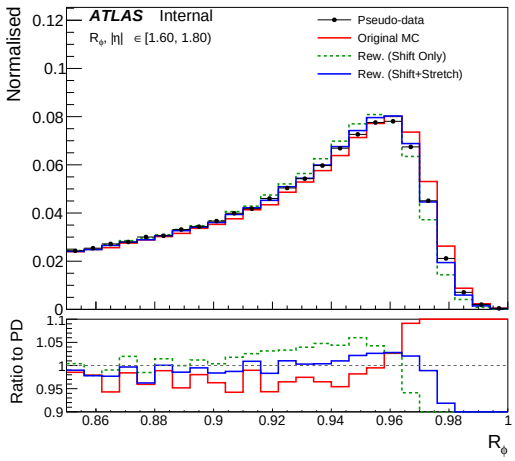


(b) R_ϕ

Figure 12: Closure plots for $|\eta| \in [1.52, 1.60)$, $\delta = 0.05$.

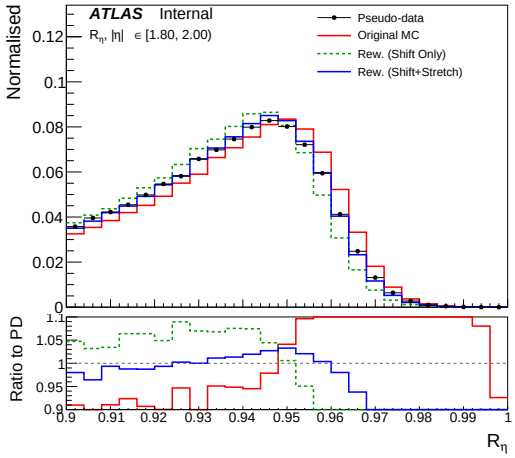


(a) R_η

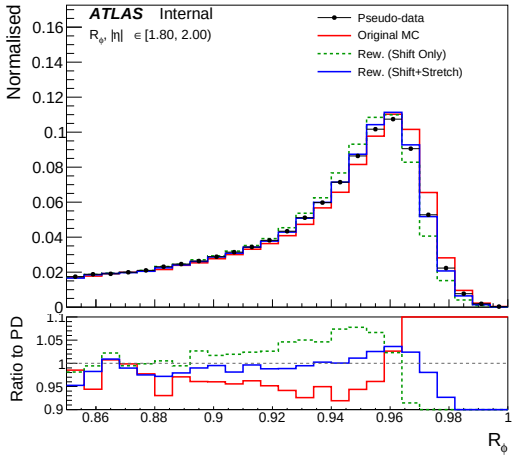


(b) R_ϕ

Figure 13: Closure plots for $|\eta| \in [1.60, 1.80)$, $\delta = 0.05$.

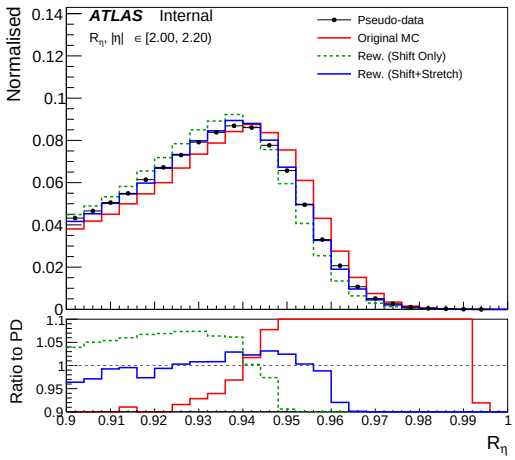


(a) R_η

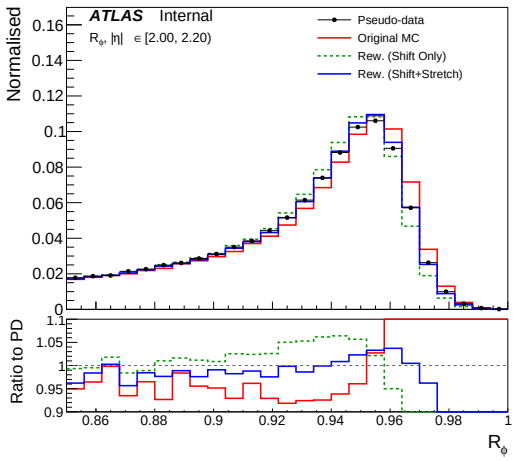


(b) R_ϕ

Figure 14: Closure plots for $|\eta| \in [1.80, 2.00)$, $\delta = 0.05$.

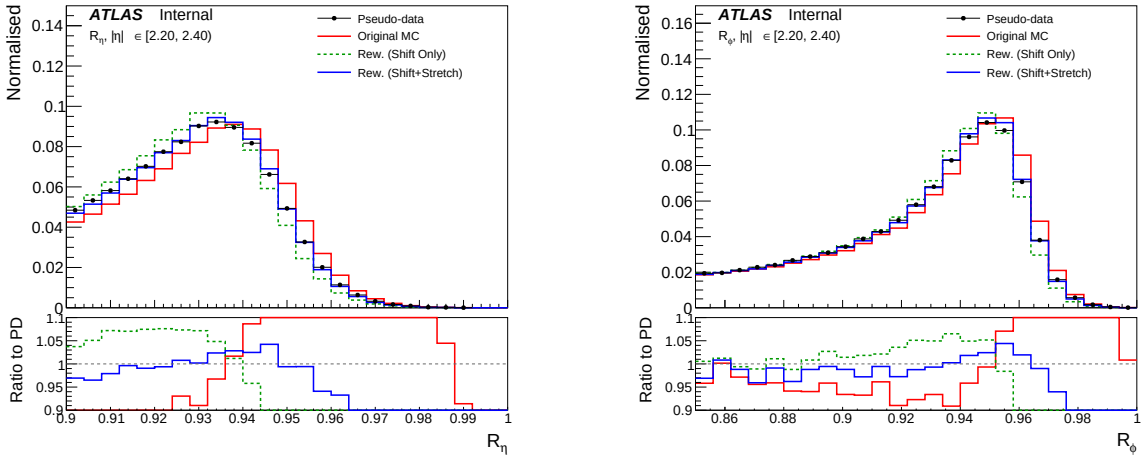


(a) R_η



(b) R_ϕ

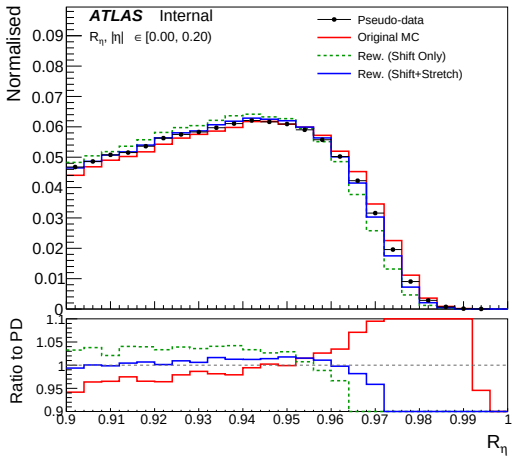
Figure 15: Closure plots for $|\eta| \in [2.00, 2.20)$, $\delta = 0.05$.



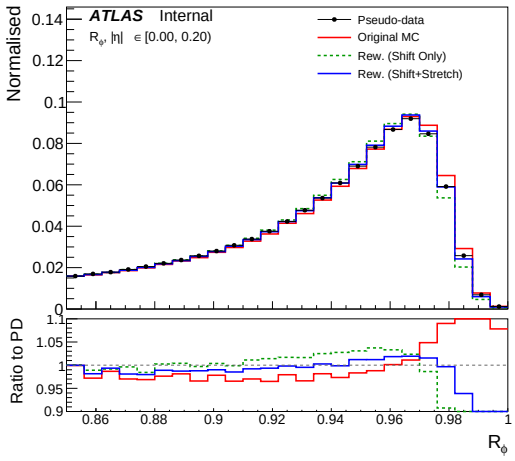
(a) R_η (b) R_ϕ

Figure 16: Closure plots for $|\eta| \in [2.20, 2.40)$, $\delta = 0.05$.

A.2 3% distortion ($\delta = 0.03$)

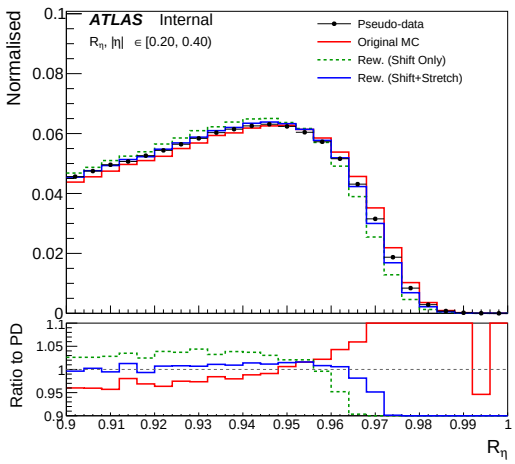


(a) R_η

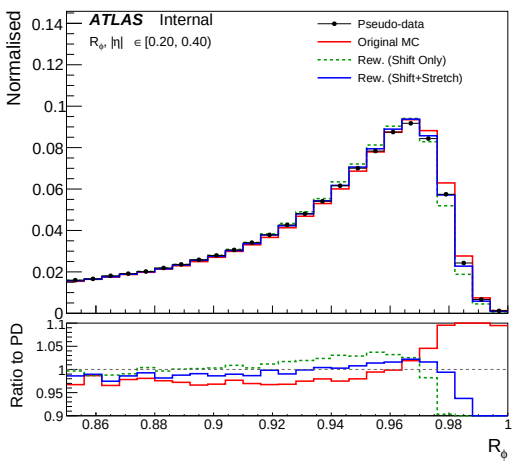


(b) R_ϕ

Figure 17: Closure plots for $|\eta| \in [0.00, 0.20)$, $\delta = 0.03$.

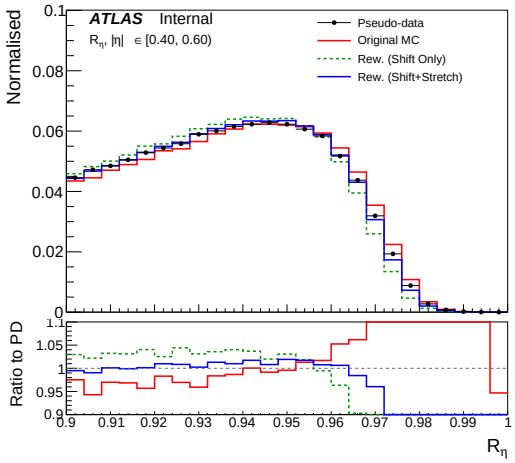


(a) R_η

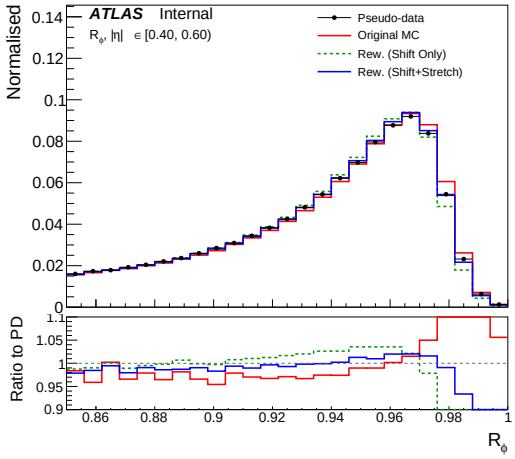


(b) R_ϕ

Figure 18: Closure plots for $|\eta| \in [0.20, 0.40)$, $\delta = 0.03$.

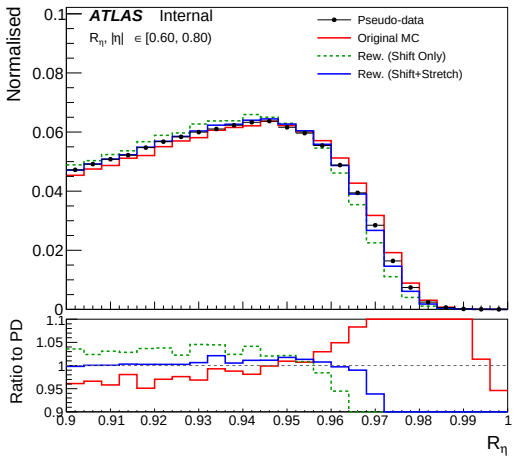


(a) R_η

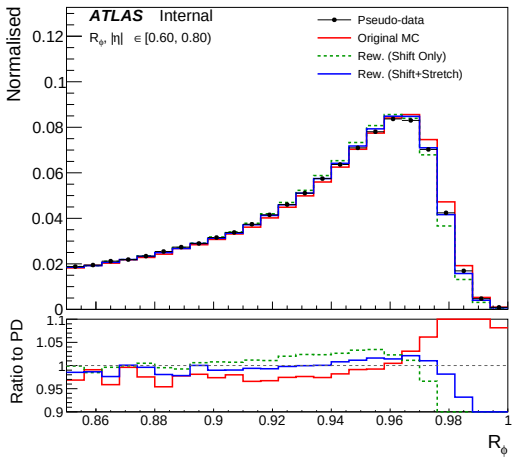


(b) R_ϕ

Figure 19: Closure plots for $|\eta| \in [0.40, 0.60)$, $\delta = 0.03$.

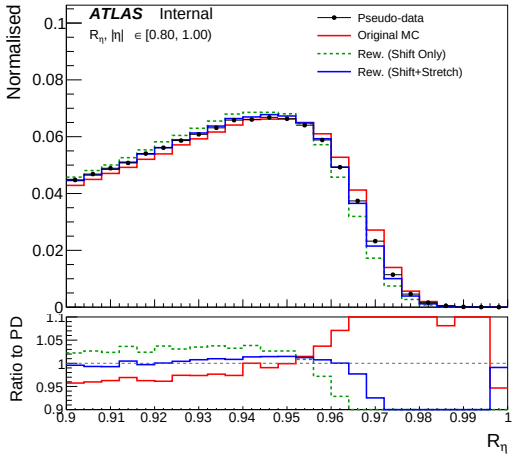


(a) R_η

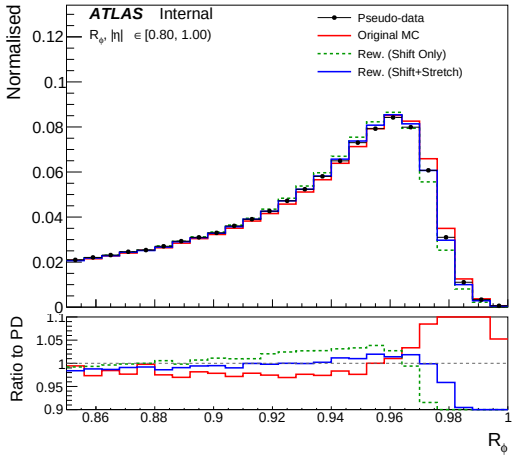


(b) R_ϕ

Figure 20: Closure plots for $|\eta| \in [0.60, 0.80)$, $\delta = 0.03$.

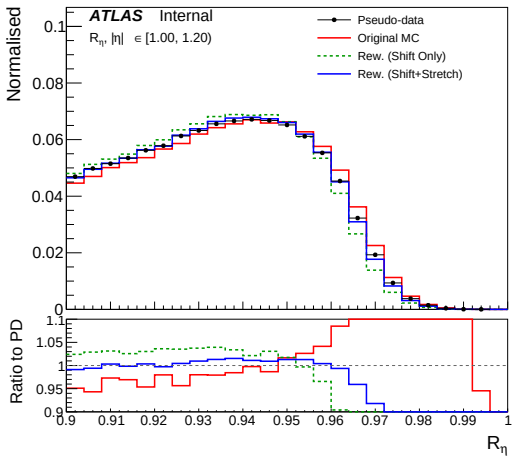


(a) R_η

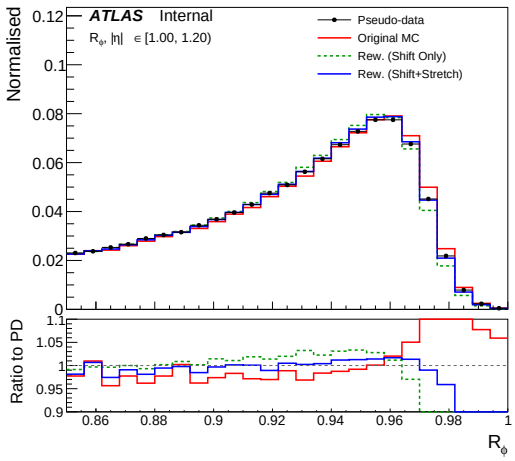


(b) R_ϕ

Figure 21: Closure plots for $|\eta| \in [0.80,\ 1.00)$, $\delta = 0.03$.

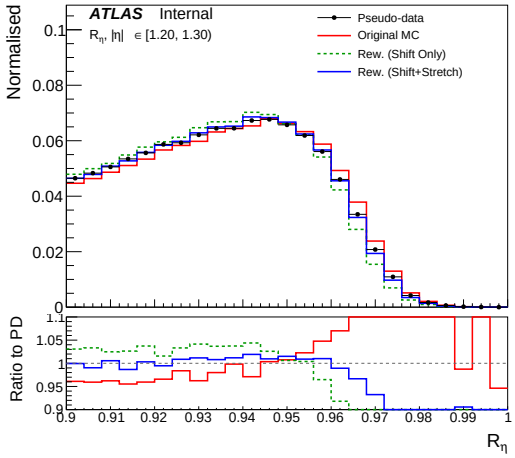


(a) R_η

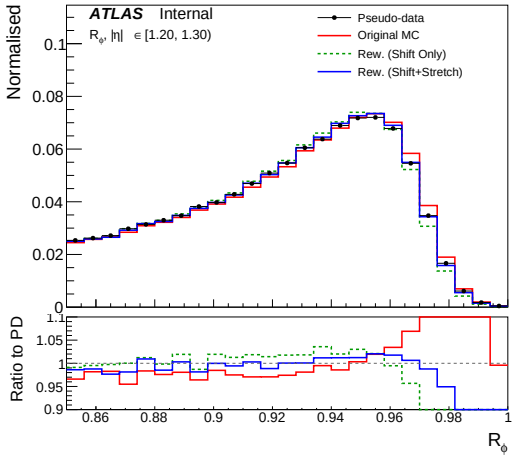


(b) R_ϕ

Figure 22: Closure plots for $|\eta| \in [1.00,\ 1.20)$, $\delta = 0.03$.

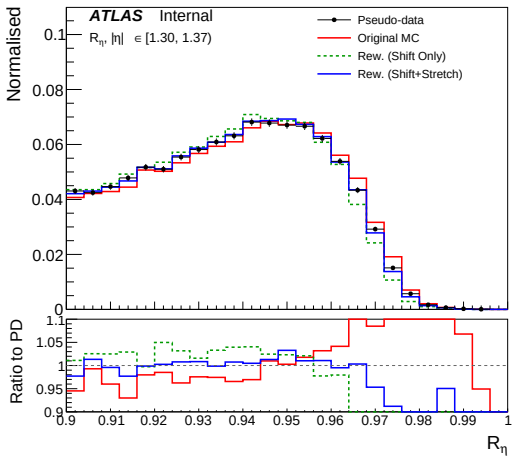


(a) R_η

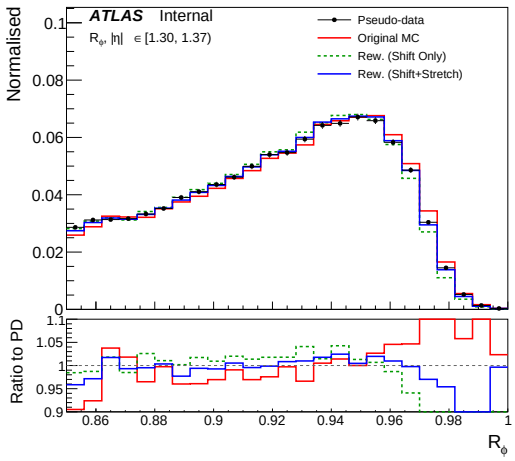


(b) R_ϕ

Figure 23: Closure plots for $|\eta| \in [1.20, 1.30)$, $\delta = 0.03$.

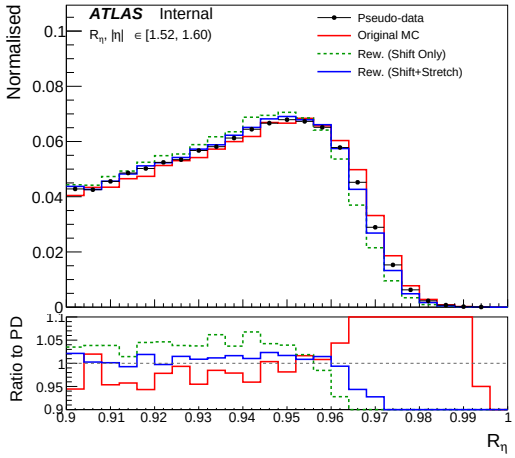


(a) R_η

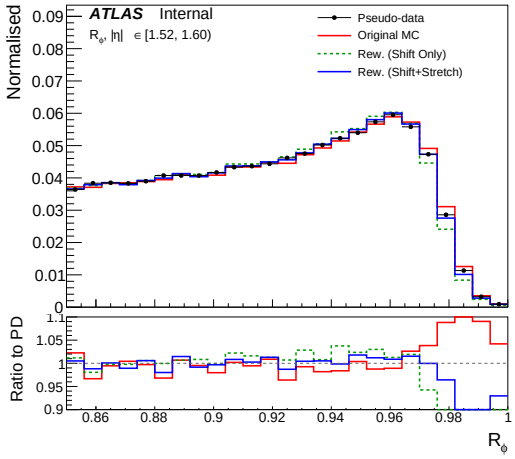


(b) R_ϕ

Figure 24: Closure plots for $|\eta| \in [1.30, 1.37)$, $\delta = 0.03$.

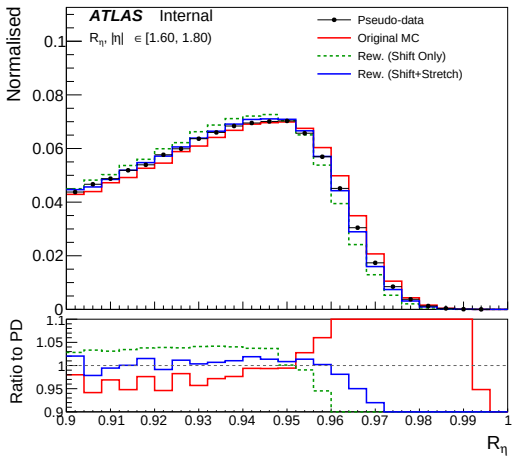


(a) R_η

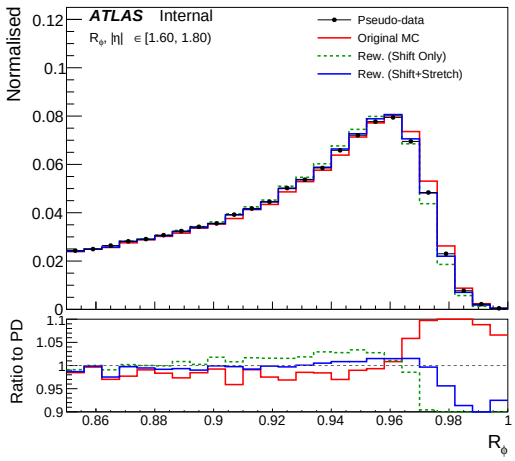


(b) R_ϕ

Figure 25: Closure plots for $|\eta| \in [1.52,\ 1.60)$, $\delta = 0.03$.

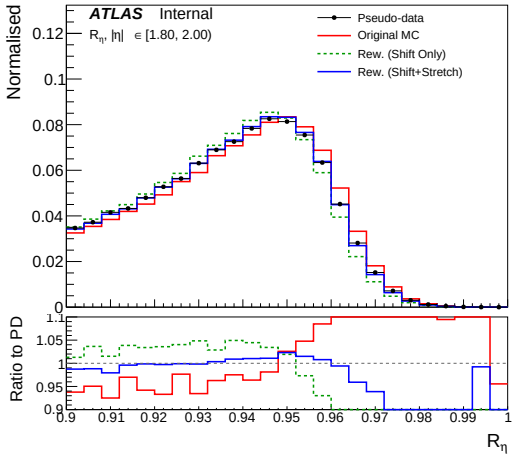


(a) R_η

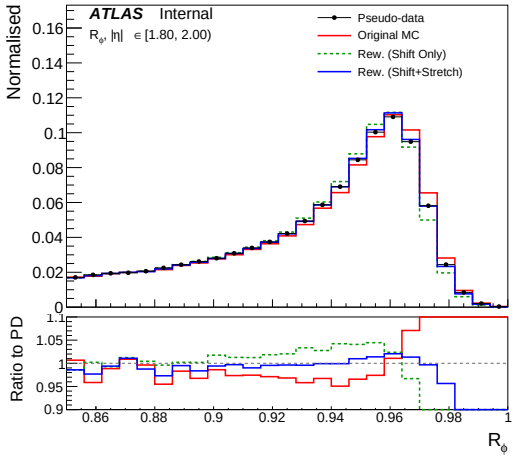


(b) R_ϕ

Figure 26: Closure plots for $|\eta| \in [1.60,\ 1.80)$, $\delta = 0.03$.

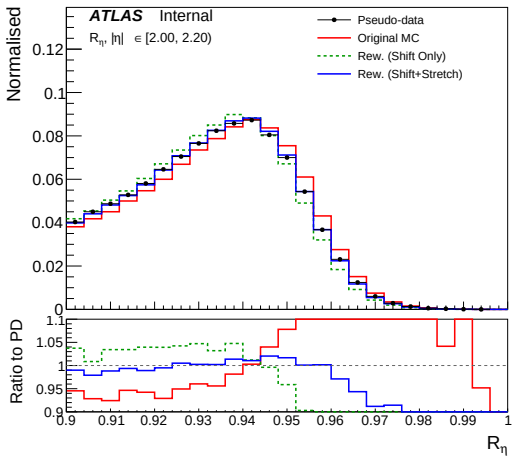


(a) R_η

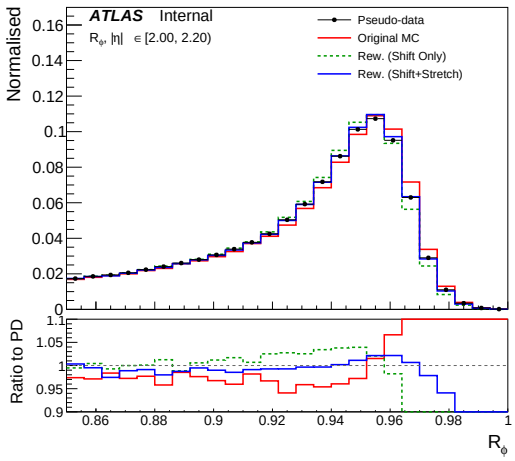


(b) R_ϕ

Figure 27: Closure plots for $|\eta| \in [1.80, 2.00)$, $\delta = 0.03$.

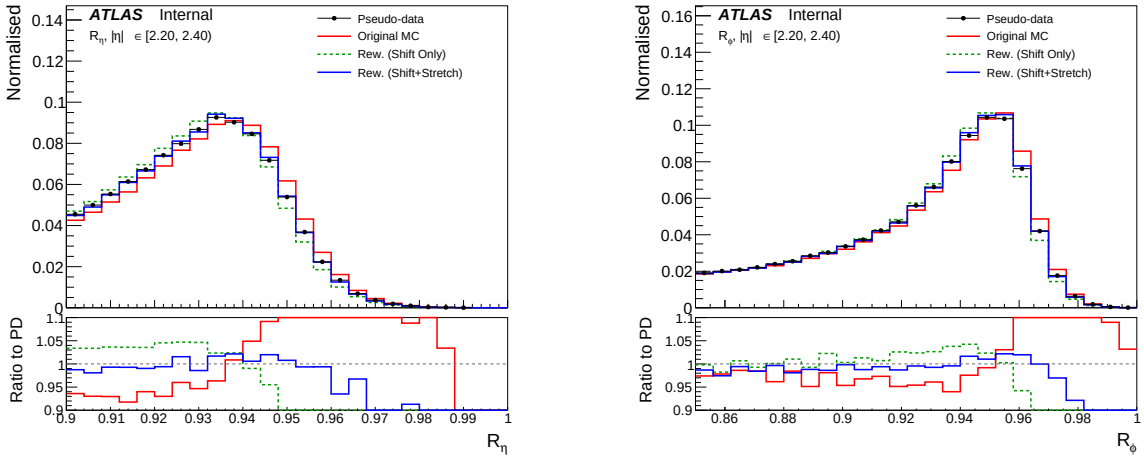


(a) R_η



(b) R_ϕ

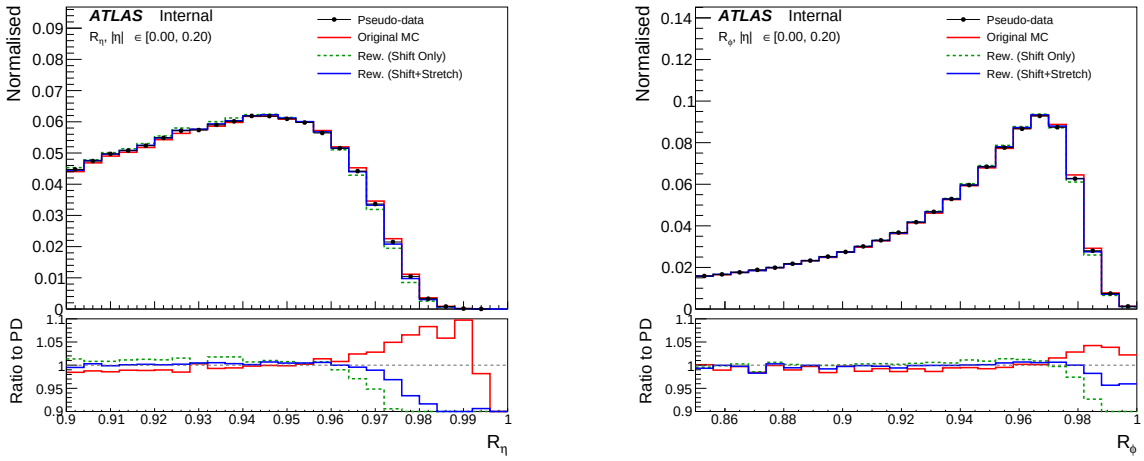
Figure 28: Closure plots for $|\eta| \in [2.00, 2.20)$, $\delta = 0.03$.



(a) R_η (b) R_ϕ

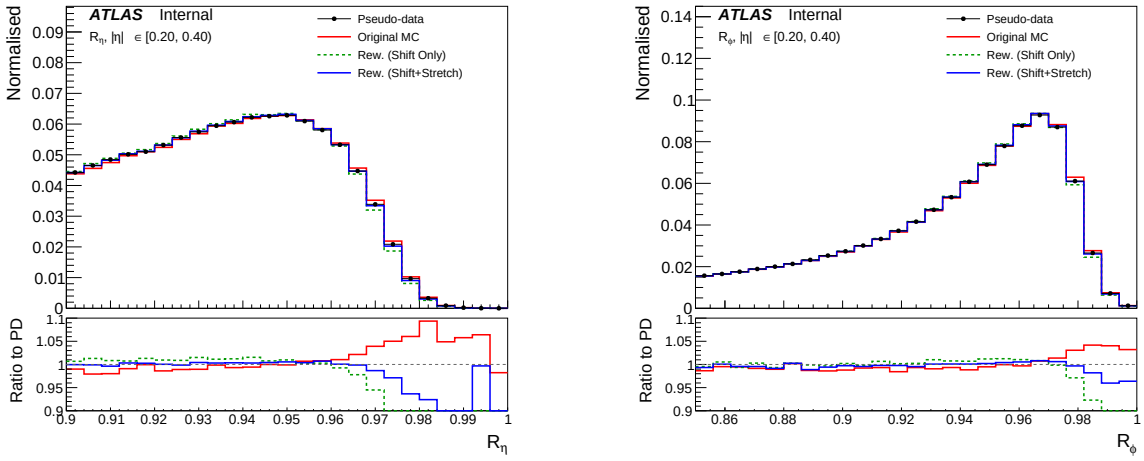
Figure 29: Closure plots for $|\eta| \in [2.20, 2.40)$, $\delta = 0.03$.

A.3 1% distortion ($\delta = 0.01$)



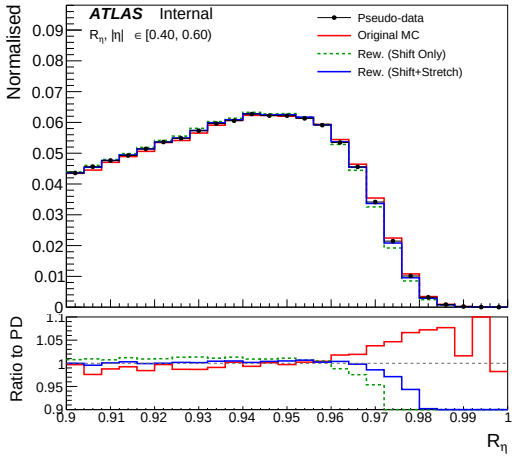
(a) R_η (b) R_ϕ

Figure 30: Closure plots for $|\eta| \in [0.00, 0.20)$, $\delta = 0.01$.

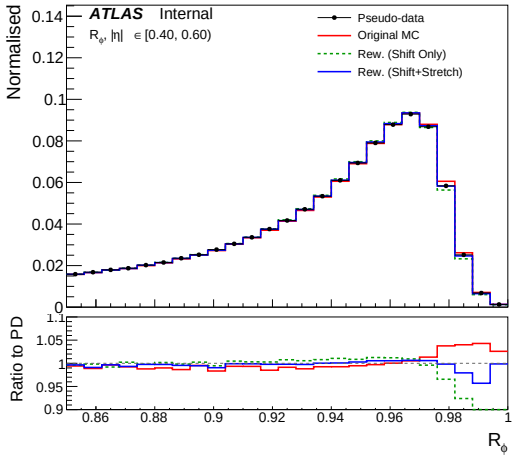


(a) R_η (b) R_ϕ

Figure 31: Closure plots for $|\eta| \in [0.20, 0.40)$, $\delta = 0.01$.

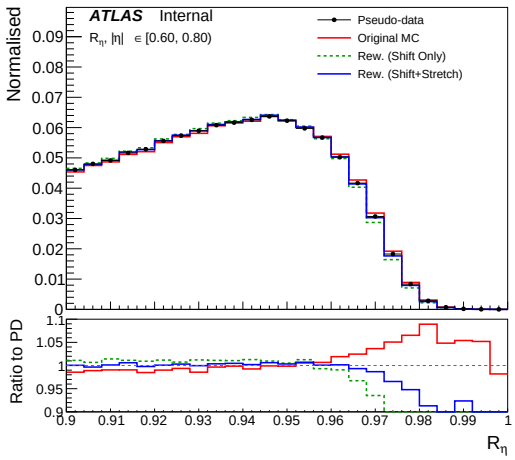


(a) R_η

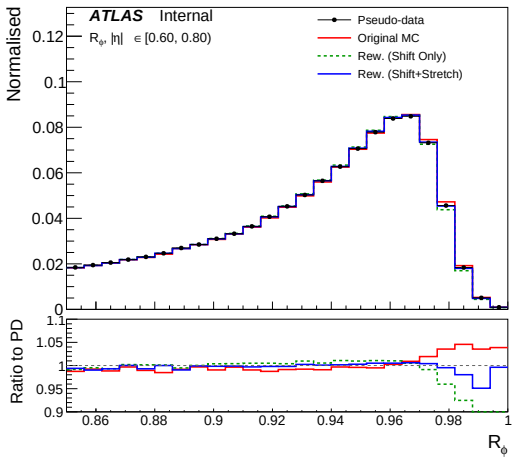


(b) R_ϕ

Figure 32: Closure plots for $|\eta| \in [0.40, 0.60)$, $\delta = 0.01$.

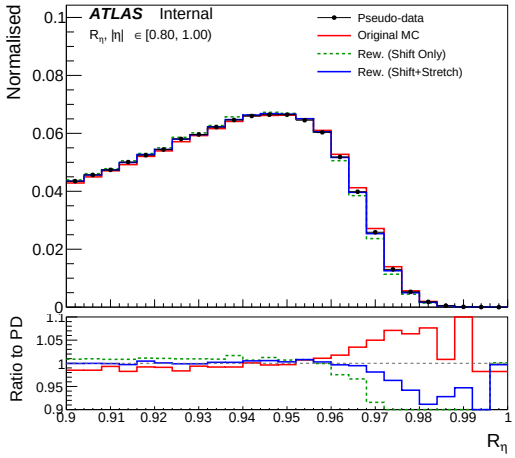


(a) R_η

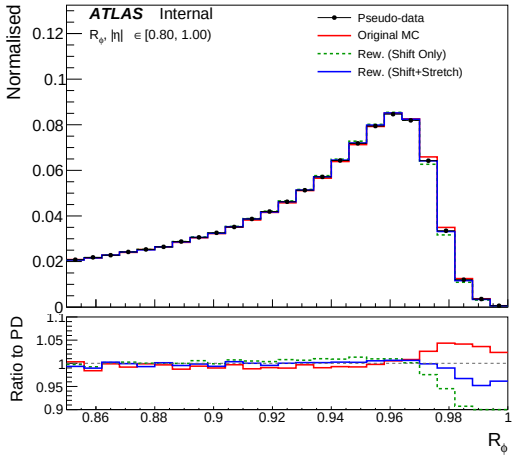


(b) R_ϕ

Figure 33: Closure plots for $|\eta| \in [0.60, 0.80)$, $\delta = 0.01$.

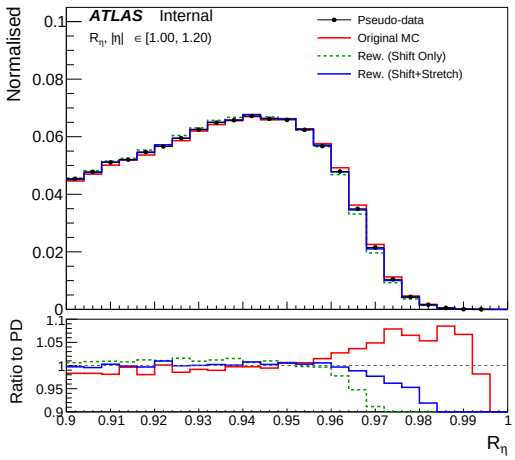


(a) R_η

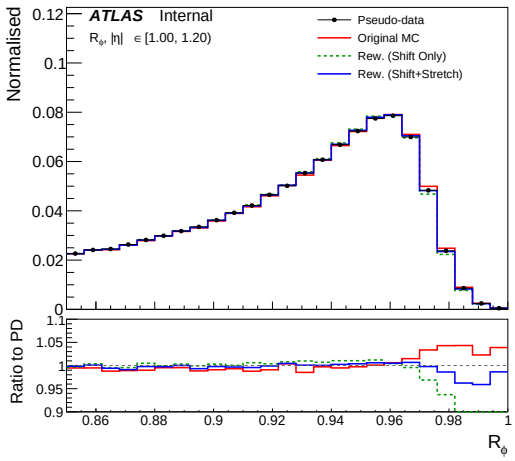


(b) R_ϕ

Figure 34: Closure plots for $|\eta| \in [0.80, 1.00)$, $\delta = 0.01$.

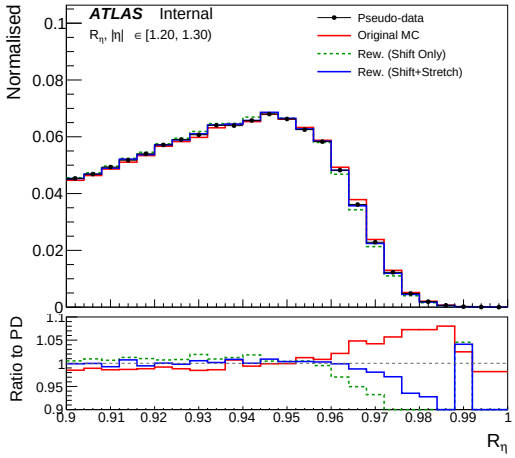


(a) R_η

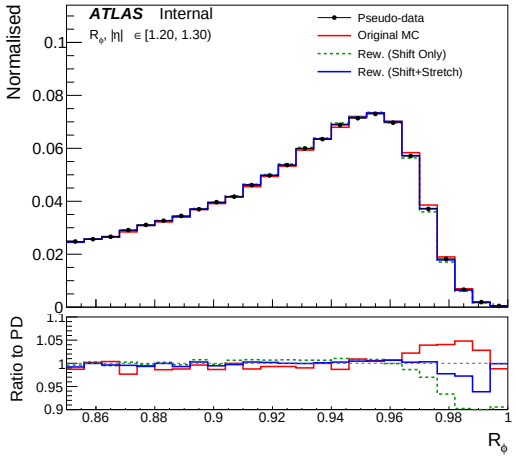


(b) R_ϕ

Figure 35: Closure plots for $|\eta| \in [1.00, 1.20)$, $\delta = 0.01$.

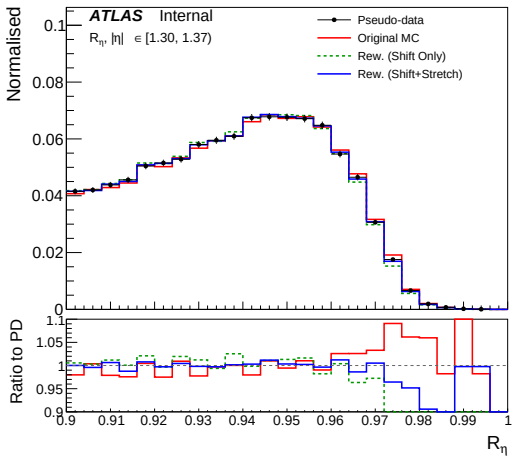


(a) R_η

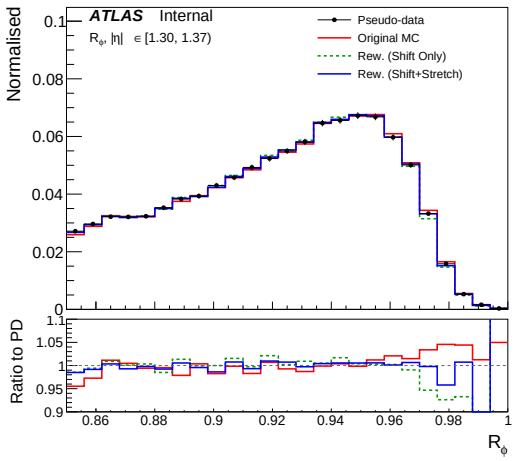


(b) R_ϕ

Figure 36: Closure plots for $|\eta| \in [1.20, 1.30)$, $\delta = 0.01$.

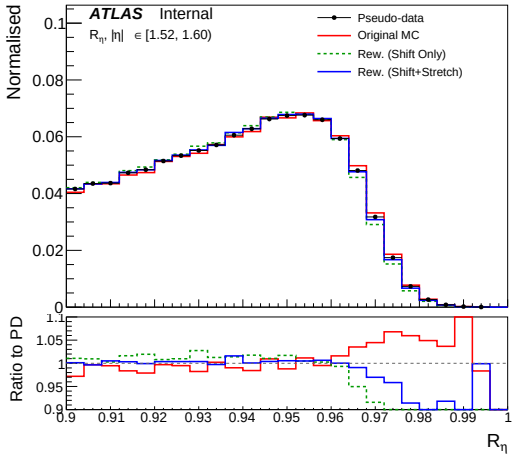


(a) R_η

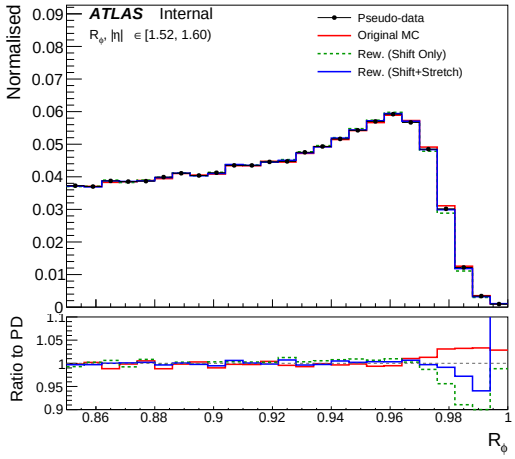


(b) R_ϕ

Figure 37: Closure plots for $|\eta| \in [1.30, 1.37)$, $\delta = 0.01$.

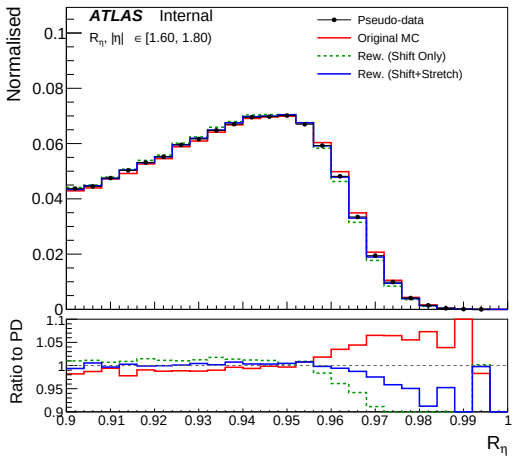


(a) R_η

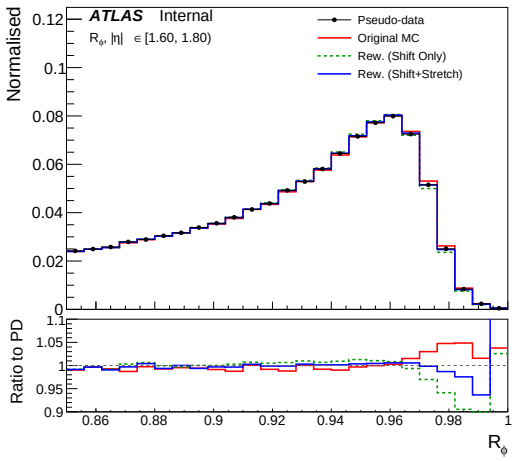


(b) R_ϕ

Figure 38: Closure plots for $|\eta| \in [1.52, 1.60)$, $\delta = 0.01$.

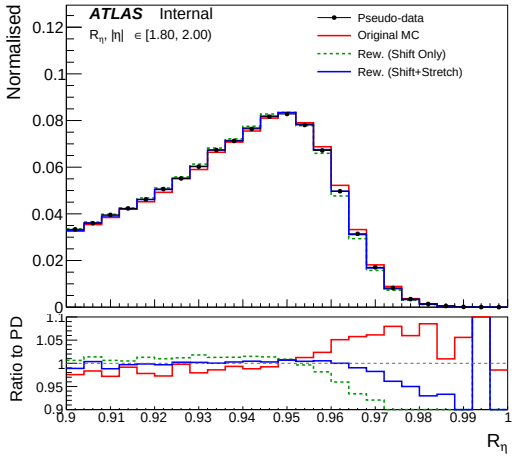


(a) R_η

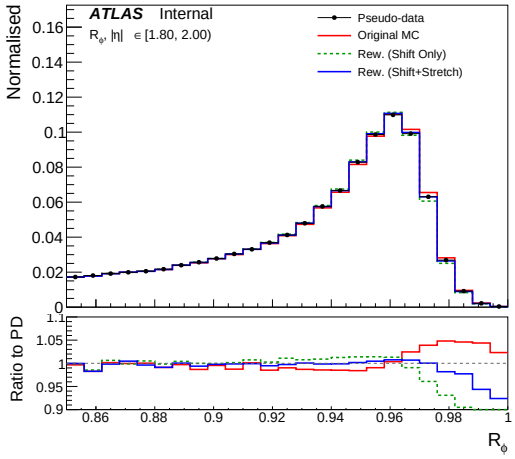


(b) R_ϕ

Figure 39: Closure plots for $|\eta| \in [1.60, 1.80)$, $\delta = 0.01$.

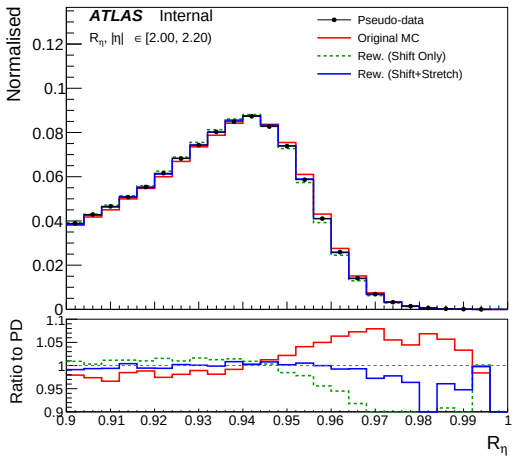


(a) R_η

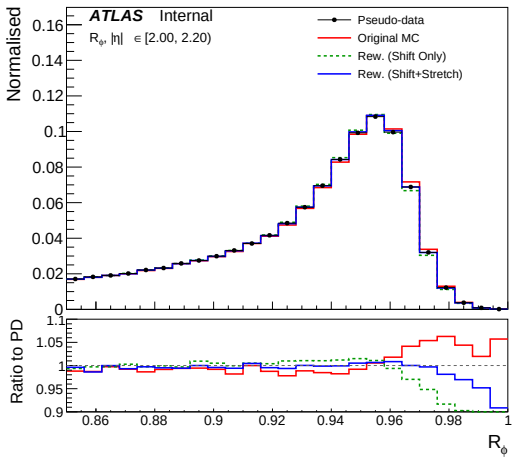


(b) R_ϕ

Figure 40: Closure plots for $|\eta| \in [1.80, 2.00)$, $\delta = 0.01$.

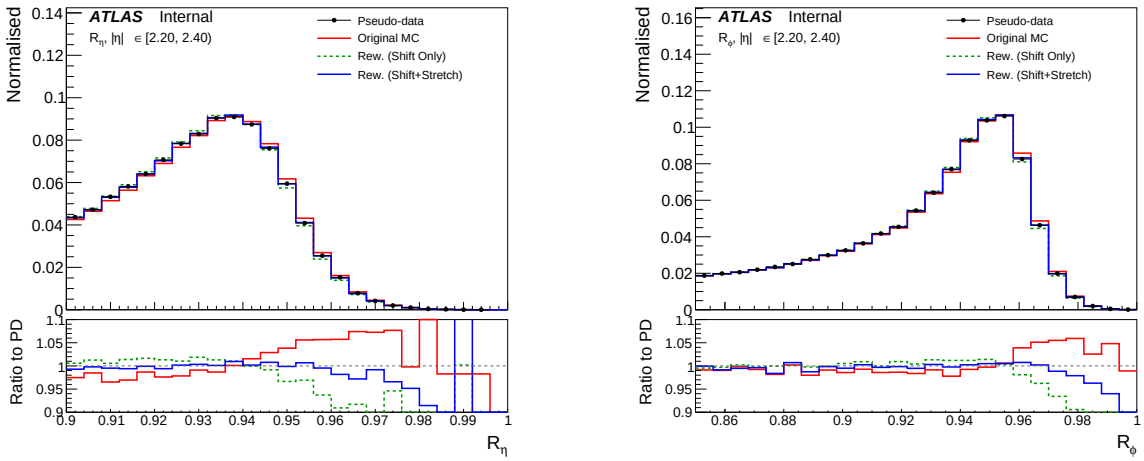


(a) R_η



(b) R_ϕ

Figure 41: Closure plots for $|\eta| \in [2.00, 2.20)$, $\delta = 0.01$.



(a) R_η (b) R_ϕ

Figure 42: Closure plots for $|\eta| \in [2.20, 2.40)$, $\delta = 0.01$.